An interior modular zero of a weight-36 Hecke eigenform, and a degree-six modular-zero polynomial

DULA
CLAUDE 4.8

June 30, 2026

Abstract

We record a concrete, fully verified object: a weight-36 level-one Hecke cusp eigenform whose two zeros in the upper half-plane form a complex-conjugate pair in the *interior* of the standard fundamental domain—off the arc $|\tau| = 1$ and off every reflection line. The six zeros of the three eigenforms of S_{36} are exactly the roots of a single irreducible degree-six polynomial $P \in \mathbb{Z}[X]$, obtained by eliminating the Hecke eigenvalue from the per-form "zero-quadratic." All displayed identities are exact and machine-checkable. We ask whether this packaging is known.

1 Setup

Let $S_{36} = S_{36}(\mathrm{SL}_2(\mathbb{Z}))$, of dimension three, and let $j = E_4^3/\Delta$ be the modular invariant ($j(i) = 1728$, $j(\rho) = 0$, $\rho = e^{2\pi i/3}$). Since $M_{24} = \langle E_4^6, E_4^3\Delta, \Delta^2\rangle = \Delta^2\langle j^2, j, 1\rangle$ and $S_{36} = \Delta \cdot M_{24}$, every $f \in S_{36}$ has the shape

$$f = \Delta^3\left(x_1\, j^2 + x_2\, j + x_3\right), \qquad (x_1, x_2, x_3) \in \mathbb{C}^3. \tag{1}$$

As Δ is nonvanishing on $\mathbb{H}$ and f vanishes to order one at the cusp, the two zeros of f in $\mathbb{H}$ lie at the points τ for which $j(\tau)$ is a root of the *zero-quadratic* $x_1X^2 + x_2X + x_3$.

The Hecke operator T_2 acts on S_{36}. In the basis $b_1 = \Delta E_4^6,\ b_2 = \Delta^2 E_4^3,\ b_3 = \Delta^3$ (equivalently $j^2\Delta^3$, $j\Delta^3$, Δ^3; leading q-orders $1, 2, 3$) it is the integer matrix with characteristic polynomial

$$\chi(x) = x^3 - 139656\, x^2 - 59208339456\, x - 1467625047588864, \tag{2}$$

irreducible over $\mathbb{Q}$ with $\operatorname{disc}(\chi) > 0$; thus the Hecke field $K = \mathbb{Q}[x]/(\chi)$ is the totally real cubic field of S_{36} (LMFDB newform `1.36.a`). For an eigenvalue θ (so $a_2 = \theta$) the eigenform's coordinates in (1) may be taken to be the cofactor column of $T_2 - \theta I$,

$$\begin{aligned}
x_1(\theta) &= \theta^2 - 138240\,\theta - 24774967296,\\
x_2(\theta) &= 34629120000\,\theta + 1502706401280000,\\
x_3(\theta) &= -17915904000000\,\theta - 8902341033984000000,
\end{aligned} \tag{3}$$

polynomials of degree ≤ 2 in θ.

2 The modular-zero polynomial

Eliminating the eigenvalue collects all modular zeros into one polynomial.

Proposition 1. *With χ and x_i as in* (2)–(3)*, set*

$$P(X) = \mathrm{Res}_\theta\big(\chi(\theta),\, x_1(\theta)X^2 + x_2(\theta)X + x_3(\theta)\big) \big/ (\textit{leading coeff}).$$

Then $P \in \mathbb{Z}[X]$ is given by

$$\begin{aligned} P(X) = {} & X^6 + 135408\,X^5 - 59796227520\,X^4 + 1622049586606080\,X^3 \\ & - 22503506634178560000\,X^2 - 85816228467179520000000\,X \\ & + 366808372441251840000000000, \end{aligned}$$

is irreducible over $\mathbb{Q}$, and has $\mathrm{disc}(P) < 0$*. Its six roots are exactly the j-invariants of the upper-half-plane zeros of the three Hecke eigenforms of S_{36}.*

All assertions in Proposition 1 are exact and verified by symbolic computation (integer Hecke matrix, symbolic resultant, irreducibility test); see §5. Because P is irreducible, the six modular-zero j-values form a single Galois orbit of degree six, and $\mathrm{disc}(P) < 0$ forces exactly one complex-conjugate pair among them. Numerically the roots are the four real values

$$-331296.33, \qquad -1398.56, \qquad 1139.16, \qquad 167923.73,$$

(zeros on the boundary $\mathrm{Re}\,\tau = \pm\frac{1}{2}$, the arc $|\tau| = 1$, and the imaginary axis, according as $j < 0$, $0 \le j \le 1728$, or $j > 1728$), together with the complex pair $14112.00 \pm 14652.78\,i$.

3 The interior zero

The complex pair belongs to the eigenform of smallest Hecke eigenvalue, $a_2 \approx -26808.0076$ (the middle root of χ): for this real embedding the discriminant $x_2^2 - 4x_1x_3$ of the zero-quadratic is negative, so its j-roots are non-real. The corresponding zero in $\mathbb{H}$ is the conjugate pair

$$\tau_0 = -0.1323718666\ldots + 1.5748742500\ldots\, i, \qquad -\overline{\tau_0},$$

with

$$j(\tau_0) = 14112.0038\ldots + 14652.7835\ldots\, i, \qquad |\tau_0| = 1.58043\ldots > 1, \quad |\,\mathrm{Re}\,\tau_0| < \tfrac{1}{2}.$$

Hence τ_0 lies in the *interior* of the standard fundamental domain. That $f(\tau_0) = 0$ is exact, by (1), since $j(\tau_0)$ is a root of the zero-quadratic.

For contrast: the weight-24 and weight-28 cusp eigenforms each carry a single *free* zero confined to the real-j locus (it recodes a_2, via $j = 696 - a_2$ and $j = 936 - a_2$ respectively; weight 28 also has a zero forced to ρ by the E_4-factor of M_{16}). Weight 36 is the first weight $\equiv 0 \pmod{12}$ at which a cusp eigenform has two genuinely free zeros, and here one Galois orbit escapes to the interior.

Proposition 2 (Threshold)**.** *For a Hecke eigenform $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$, the zeros of f in $\mathbb{H}$ other than those forced to ρ or i are the points τ with $j(\tau)$ a root of a* free-zero polynomial *Q_f of degree* $\dim S_k - 1$*, whose coefficients lie in the (totally real) Hecke field. An interior zero, i.e. one with $j(\tau) \notin \mathbb{R}$, therefore requires* $\deg Q_f \ge 2$*, that is* $\dim S_k \ge 3$*. The least weight with* $\dim S_k \ge 3$ *is* $k = 36$*. Hence weight* 36 *is the smallest weight admitting an interior cusp-form zero, and the pair* $\{\tau_0, -\overline{\tau_0}\}$ *above realises it.*

4 Questions

The construction is elementary, and zeros of modular forms are classically studied: the zeros of E_k lie on the arc $|\tau| = 1$ (Rankin–Swinnerton-Dyer), while the zeros of Hecke cusp eigenforms equidistribute on the modular surface with respect to the hyperbolic measure as $k \to \infty$ (Rudnick). Since the arc has measure zero, interior zeros are in fact *generic* at large weight; the interest of weight 36 is that it is the *threshold* (Proposition 2). My questions concern the specific packaging.

Question 1. Is $P(X) = \mathrm{Res}_\theta(\chi,\ \text{zero-quadratic})$—the "modular-zero resultant" of the Hecke basis—a named or previously studied object? The construction $P_k = \mathrm{Res}_\theta(\chi_k, Q_k)$ is uniform in k (with Q_k the degree-$(\dim S_k - 1)$ free-zero polynomial); is there a closed form or generating description, e.g. across $k \equiv 0 \pmod{12}$?

Question 2. For weight 36 the criterion for an interior zero is exactly $\delta(\theta) = x_2(\theta)^2 - 4x_1(\theta)x_3(\theta) < 0$ at the embedding $a_2(f)$. For higher weight the criterion becomes "the free-zero polynomial Q_f has a non-real root." Is there a clean characterization of which eigenforms have interior zeros across weights? (Empirically it is *not* governed by the size of a_2: at weight 48 the two eigenforms with interior zeros have the smallest and the largest $|a_2|$, while the middle pair stay on the boundary.)

Question 3. The value j_0 is an algebraic integer (a root of the monic $P \in \mathbb{Z}[X]$), yet τ_0 is *not* a CM point: numerically τ_0 is not imaginary quadratic—no integer relation $a\tau_0^2 + b\tau_0 + c = 0$ exists with $\max(|a|, |b|, |c|) \le 10^6$—so by Schneider's theorem τ_0 is transcendental. Does the sextic field $\mathbb{Q}(j_0)$, or the algebraic integer j_0 itself, carry any further arithmetic structure?

5 Verification

Every displayed identity is reproduced from scratch by a short `sympy/mpmath` script: it builds the integer T_2 matrix on S_{36}, recovers χ and the cofactor coordinates (3), forms P as the resultant and checks it against the stated coefficients, certifies irreducibility and $\mathrm{disc}(P) < 0$, and locates τ_0 to high precision with an independent vanishing check. Available on request.